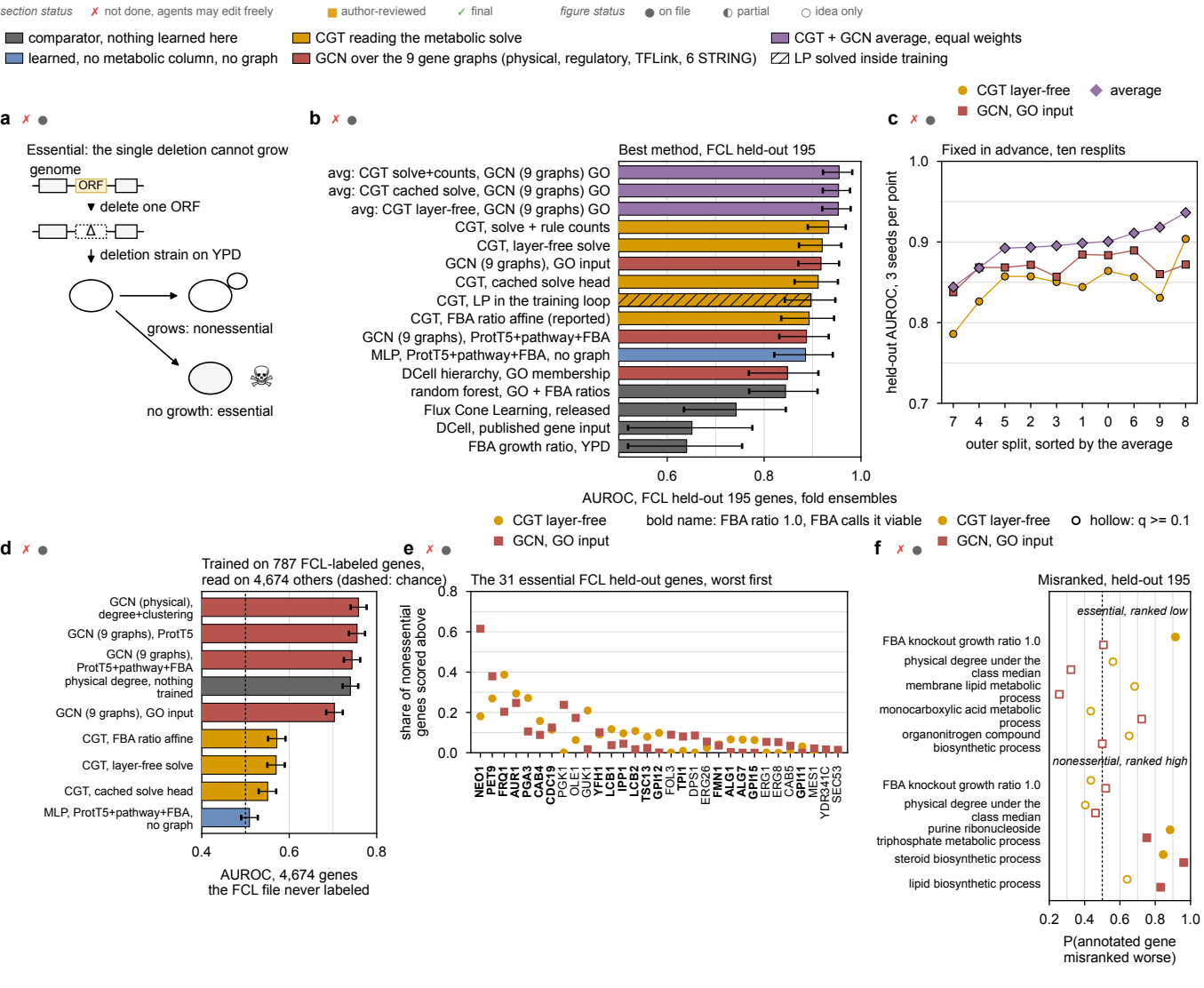

section status ✗ not done, agents may edit freely ■ author-reviewed ✓ final figure status ● on file ◐ partial ○ idea only
comparator, nothing learned here
learned, no metabolic column, no graph
CGT reading the metabolic solve
GCN over the 9 gene graphs (physical, regulatory, TFLink, 6 STRING)
CGT + GCN average, equal weights
LP solved inside training
a ✗ ●
Essential: the single deletion cannot grow
genome
ORF
delete one ORF
deletion strain on YPD
grows: nonessential
no growth: essential
b ✗ ●
Best method, FCL held-out 195
avg: CGT solve+counts, GCN (9 graphs) GO
avg: CGT cached solve, GCN (9 graphs) GO
avg: CGT layer-free, GCN (9 graphs) GO
CGT, solve + rule counts
CGT, layer-free solve
GCN (9 graphs), GO input
CGT, cached solve head
CGT, LP in the training loop
CGT, FBA ratio affine (reported)
GCN (9 graphs), ProtT5+pathway+FBA
MLP, ProtT5+pathway+FBA, no graph
DCell hierarchy, GO membership
random forest, GO + FBA ratios
Flux Cone Learning, released
DCell, published gene input
FBA growth ratio, YPD
0.6
0.8
1.0
AUROC, FCL held-out 195 genes, fold ensembles
CGT layer-free
GCN, GO input
bold name: FBA ratio 1.0, FBA calls it viable
c ✗ ●
CGT layer-free
average
GCN, GO input
Fixed in advance, ten resplits
held-out AUROC, 3 seeds per point
1.0
0.9
0.8
0.7
7 4 5 2 3 1 0 6 9 8
outer split, sorted by the average
d ✗ ●
Trained on 787 FCL-labeled genes,
read on 4,674 others (dashed: chance)
GCN (physical), degree+clustering
GCN (9 graphs), ProtT5
GCN (9 graphs), ProtT5+pathway+FBA physical degree, nothing trained
GCN (9 graphs), GO input
CGT, FBA ratio affine
CGT, layer-free solve
CGT, cached solve head
MLP, ProtT5+pathway+FBA, no graph
0.4
0.6
0.8
AUROC, 4,674 genes the FCL file never labeled
e ✗ ●
The 31 essential FCL held-out genes, worst first
share of nonessential genes scored above
0.8
0.6
0.4
0.2
0.0
NEO1 PET9 FRQ1 AUR1 PGA3 CAB4 CDC19 PGK1 OLE1 GUK1 YPH1 LCB1 IPP1 LCB2 TSC12 GPI12 FOL3 TPI1 DPS1 ERG26 FMN1 ALG1 ALG7 GPI15 ERG1 ERG8 CAB5 GPI11 MES1 YDR341C SEC53
CGT layer-free
GCN, GO input
hollow: q >= 0.1
f ✗ ●
CGT layer-free
GCN, GO input
Misranked, held-out 195
essential, ranked low
FBA knockout growth ratio 1.0
physical degree under the class median membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
nonessential, ranked high
FBA knockout growth ratio 1.0
physical degree under the class median purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
0.2
0.4
0.6
0.8
1.0
P(annotated gene misranked worse)